

LESSON 4.5: HealthCheck Utility

What this lesson is about

This lesson describes about HealthCheck user interface and importance of using HealthCheck tool. Also explains about scripts of Sterling B2B Integrator system.

What you should be able to do

After completing this lesson, you should be able to:

- Explain about HealthCheck tool.
- Discuss the importance of HealthCheck tool.
- Explain the procedure to install HealthCheck tool.
- Describe the steps to customize HealthCheck tool.
- Explain the importance of the scripts in Sterling B2B Integrator.

How you will check your progress

- The progress of the lesson is analyzed based on the successful application of the topics in the scenario and test the exercise independently.

References

Documentation

- HealthCheck:
http://www.ibm.com/support/knowledgecenter/SS3JSW_5.2.0/com.ibm.help.healthcheck.doc/com.ibm.help.healthcheck.welcome.doc/healthcheck_welcome.html
-

Overview

Introduction

HealthCheck is a utility package with Sterling B2B Integrator that provides system diagnostics, error analysis and reporting to the Sterling B2B Integrator users.

HealthCheck scanners test on Sterling B2B Integrator components such as business process configuration, system logs, service scheduling, Java Virtual Machine, database server, and operating system are optimally configured for Sterling B2B Integrator.

HealthCheck Scanners

Scanners are collections of tests and sub-tests that are grouped by function. This test ensures optimization in multiple components of Sterling B2B Integrator. Following are the scanner components the scanners:

- Business Process Scanner
- Config Scanner
- Database Scanner
- Dynamic Tracking Data
- Host System Scanner
- Log Scanner
- Password Scanner
- Schedule Scanner
- Static Config Scanner
- System Process Scanner

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Overview

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HealthCheck Scanners

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For Example, The Host System Scanner ensures that the OS-settings are optimized for Sterling B2B Integrator. The Host Scanner comprises the following tests:

- Dump Files
- Disk Free
- Document Directory Count
- Old Logs
- Thread Count
- ULimit
- CPU Used
- IO Wait
- Processes
- NFS Mount

For more details on other types of scanner, refer

http://www.ibm.com/support/knowledgecenter/SS3JSW_5.2.0/com.ibm.help.healthcheck.doc/com.ibm.help.healthcheck.reference.doc/healthcheck_scanners.html

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Overview

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Enhancements in HealthCheckv1.0.3.1

The enhancements in v1.0.3.1 are:

- User interface and 10x.jar are updated to access HealthCheck from the browser.
- Password scanner ensures the strength of user's passwords is secured and consists of two user test, one super user test, and the all user test.
- Set a limit on the number of failures a particular test can report, to avoid reworking on similar errors.
- The default scanner's testTimeout property is adjusted to 120 seconds (2 minutes) and scanner timeout property for the scanner is changed to 1200 seconds(20 minutes).
- The PDF report bar graphs use an integer scale on the y-axis. The test and groups failures in a particular test is reported based on occurrences.
- Index fragmentation, table compression, and index compression test can be selected to exclude in the database scanner.

Importance of HealthCheck

Following are the importance of HealthCheck:

- HealthCheck tool can automate the system health checks and flag common errors and potential issues allow customers to self-diagnose.
 - It provides the recommendations to resolve any warnings or errors that are identified by the utility in the user interface.
 - It provides guidelines and suggested actions to help the Sterling B2B Integrator system run efficiently.
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Installation

Introduction

HealthCheck tool is automatically installed with Sterling B2B Integrator V5.2.6.1.

In earlier version, following steps were used to install HealthCheck tool:

1. Download the HealthCheck.jar from fix central.
2. Copy the HealthCheck.jar file into the <install_dir/packages directory.
3. From <install_dir>/bin , enter the command hardstop.sh on a UNIX system to stop all Sterling B2B Integrator processes.
4. Run the InstallServer script from <install_dir>/packages folder
5. InstallService.sh healthCheck_XXXXX.jar
6. where XXXXX is the version number for HealthCheck.
7. From <install_dir>/bin directory, enter the run.sh on a UNIX system to start all Sterling B2B Integrator processes.

Access HealthCheck

Access the HealthCheck user interface through the browser. From the <install_dir>/bin directory run healthcheck.sh on UNIX system to view HealthCheck user interface. OR

From the browser access http://SI_Install_IP_address:portnumber/healthcheck.



Example

<http://192.198.10.100:9000/healthcheck>.

Exercise 4.5.1: Access HealthCheck User Interface

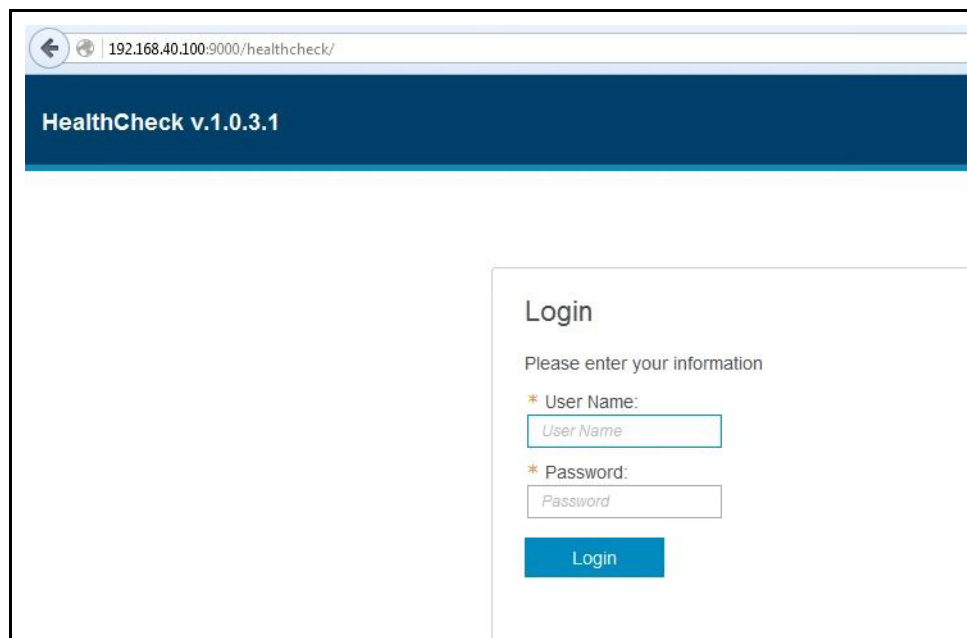
Introduction

Angela want to access the default HealthCheck user interface to further work on the tool to use the importance of the tool.

Instructions

Complete the following steps to access the user interface.

Step 1: Access the windows image, open a web browser to access **http://192.168.40.100:9000/healthcheck** URL.

A screenshot of a web browser window showing the HealthCheck v.1.0.3.1 login page. The browser's address bar displays the URL '192.168.40.100:9000/healthcheck/'. The page has a dark blue header with the text 'HealthCheck v.1.0.3.1'. Below the header, there is a white box containing a 'Login' form. The form includes the text 'Please enter your information', followed by two fields: '* User Name:' with a text input box containing 'User Name', and '* Password:' with a text input box containing 'Password'. At the bottom of the form is a blue 'Login' button.

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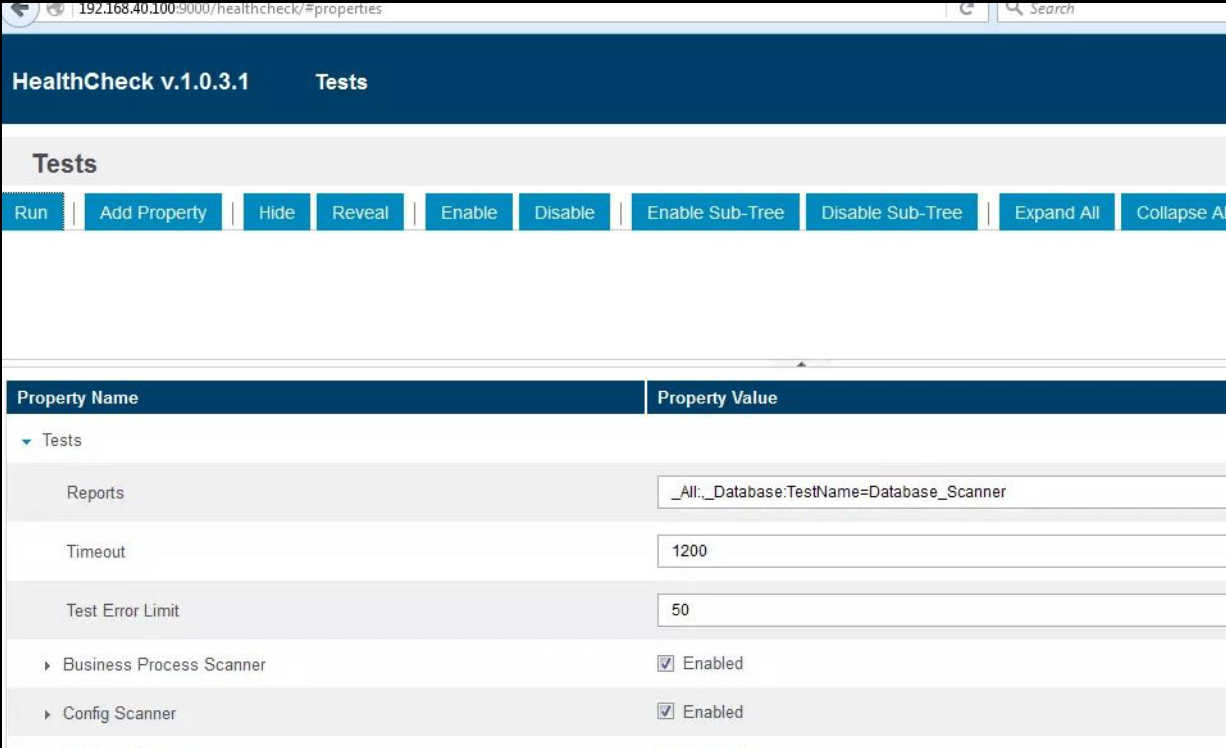
Exercise 4.5.1: Access HealthCheck User Interface

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Instructions

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Step 2: To view the HealthCheck user interface, specify **admin** as the Username and **password** as the Password.



HealthCheck v.1.0.3.1 Tests

Tests

Run | Add Property | Hide | Reveal | Enable | Disable | Enable Sub-Tree | Disable Sub-Tree | Expand All | Collapse All

Property Name	Property Value
▼ Tests	
Reports	_All:._Database:TestName=Database_Scanner
Timeout	1200
Test Error Limit	50
▶ Business Process Scanner	☒ Enabled
▶ Config Scanner	☒ Enabled

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Exercise 4.5.1: Access HealthCheck User Interface

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Instructions

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Complete the following steps to run the report by using the default property settings in the user interface:

Step 1: From the healthcheck user interface, click **Run** tab to create the report. The screen capture depicts the run status.

The screenshot shows the 'Run' tab of the HealthCheck User Interface. On the left, there is a log of test results for the 'Config Scanner' and 'Static Config Scanner'. On the right, a summary table titled 'Testing Complete!' shows the status of various scanners and the number of errors found.

Status	Scanner	Results
✓	Config Scanner	3 errors
✓	Static Config Scanner	0 errors
✓	Host System Scanner	1 errors
✓	Database Scanner	14 errors
✓	Dynamic Tracking Data Scanner	0 errors
✓	Schedule Scanner	0 errors
✓	Password Scanner	37 errors
✓	System Process Scanner	0 errors
✓	Business Process Scanner	0 errors

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Exercise 4.5.1: Access HealthCheck User Interface

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Instructions

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Step 2: Click **View Summary** tab, the report is displayed with all the details as shown in the screen capture.

The screenshot shows the HealthCheck User Interface. At the top, there are tabs: All, Database Scanner, Host System Scanner, Password Scanner, Business Process Scanner, and Config Scanner. Below the tabs is a 'Download Report' button. The 'Summary' tab is selected, showing a table of results. The table has columns for severity, check name, count, and description. The results are as follows:

Severity	Check Name	Count	Description
SEVERE	DB2 dft_mon_stmt Check	1	Change dft_mon_stmt to ON
SEVERE	DB2 dft_mon_table Check	1	Change dft_mon_table to ON
SEVERE	DB2 dft_mon_uow Check	1	Change dft_mon_uow to ON
SEVERE	Default Password Check	37	Change password to something
ERROR	Customer Overrides Check	3	Do not modify property files dire
ERROR	Java Core and Heap Dump Files Check	1	Examine and address any issue manually generated. Move or d heapdump.20160816.072443.12
WARNING	Index Fragmentation	8	Consider rebuilding the index. I reorganization, even after a table reorgchk utility suggestions and

Step 3: Click **Database Scanner** tab to view only database-related report.

Step 4: Click **Host System Scanner** tab to view only host system-related report.

Step 5: Click **Password Scanner** tab to view only password-related report.

Step 6: Click **Business Process Scanner** tab to view business process-related report.

Step 7: Click **Config Scanner** tab to view configuration-related report.

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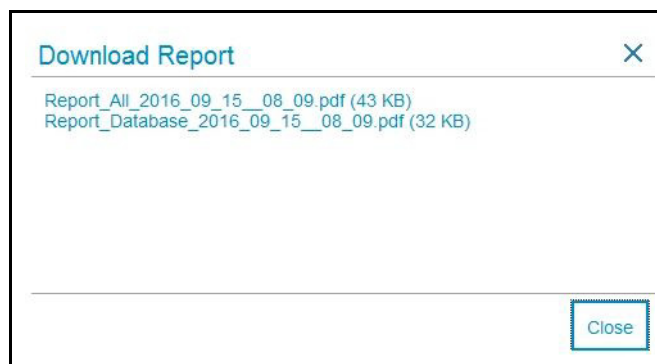
Exercise 4.5.1: Access HealthCheck User Interface

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Instructions

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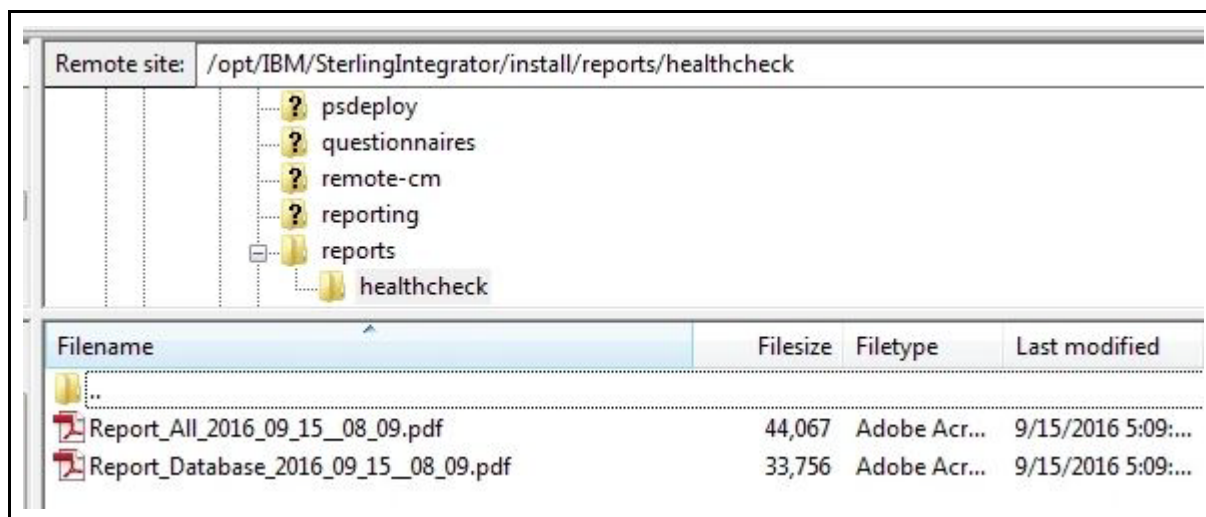
Step 8: Click **Download Report** tab, the screen capture depicts the download report.



Step 9: Click **Close**.

Step 10: From the FileZilla tool, access Sterling B2B Integrator installed directory.

Step 11: The report files are stored in the `/opt/IBM/SterlingIntegrator/install/reports/healthcheck` folder as shown in the screen capture.

**Note**

To conserve system resource, ensure that reports are regularly cleared.

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Exercise 4.5.1: Access HealthCheck User Interface

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Instructions

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Step 12: Select Report_ALL_XXXX.pdf file.

Step 13: **Right-click -> View/Edit** to view the report in the PDF format. The screen capture depicts the partial view of the report.



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Exercise 4.5.1: Access HealthCheck User Interface

(Continued)

Instructions

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Note

The severity status and color indicates as shown in the table:

Code Key	Description
1 - Warning	is in lowest level, is displayed in green in the reports and the user interface
2 - Error	is in second highest severity level, is displayed in yellow or orange.
3 - Severe	is in highest severity level, is displayed in red.
Other - Unknown	any other error, is displayed in blue

Step 14: Review the error description in the report and analyze the reasons for the performance.

User Interface

Functions

The table describes the functions of the HealthCheck user interface tabs:

Tab	Description
Run	Executes all the enables scanners and properties to generate a report.
Add Property	Used to add a property. For Example Select Timeout property, Click Add Property tab to add new sub property with new value, as shown in the screen capture.  This properties will be add in the healthcheck properties file.
Hide	Used to hide the property name from the display, so that the reports will hide the property in the report.
Reveal	Hidden property can be retrieved.
Enable	Used to enable the scanners. You need to highlight the scanner and click Enable tab.
Disable	Used to disable the scanners. You need to highlight the scanner and click Disable tab.
Enable Sub-Tree	Used to enable the sub-tree, For example, highlight DocToDOM from Business Process Scanner and select Enable Sub-Tree.
Disable Sub-Tree	Used to enable the sub-tree, For example, highlight DocToDOM from Business Process Scanner and select Disable Sub-Tree.
Expand All	Double click to expand entire properties and scanners.
Collapse All	Double click to collapse entire properties and scanners.
Expand Sub-Tree	Double click to expand selected scanner's sub-tree.
Collapse Sub-Tree	Double click to collapse selected scanner's sub-tree

HealthCheck Command

Introduction

Reports can be created and viewed through the command line interface. To configure and customize tests, extensible properties can be used.

The table depicts the list of commands that are used to run the HealthCheck utility from the command line:

Command	Description
-h	help command lists the abbreviated scanner and test names.
-ALL	enables and runs all scanners and tests in HealthCheck.
-S	scanner command enables and runs scanners with the scanner short name. For Example, -S BPS SPS, where BPS is BusinessProcess Scanner, SPS is System Process Scanner.
-T	test command enables and runs specific test with the scanner short name. For example, -T BPS.xPath, run only XPath test in the Business Process Scanner.

Exercise 4.5.2: HealthCheck Command

Introduction

Angela as an administrator runs HealthCheck of Sterling B2B Integrator components. Angela is asked to use few HealthCheck utility commands to optimize the Sterling B2B Integrator.

Angela first access the HealthCheck utility help command to know the command details.

Instructions

Complete the following steps to access the help command:

Step 1: From the Windows image, click PuTTY icon.

Step 2: Select **SI** from Save Session.

Step 3: Click **Open**.

Step 4: Enter **root** for login.

Step 5: Click **Enter**.

Step 6: Enter **root123** for password.

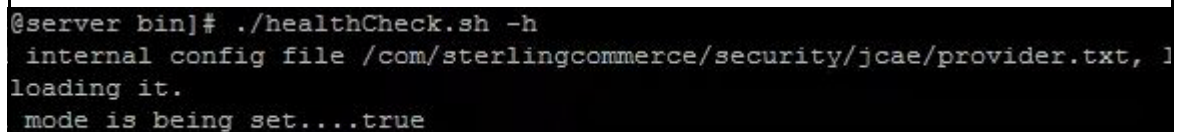
Step 7: Click **Enter**.

Step 8: Enter **cd /opt/IBM/SterlingIntegrator/install/bin**

Step 9: Click **Enter**.

Step 10: Enter **./healthCheck.sh -h**

Step 11: Click **Enter**, you see that script loads as shown in the screen capture.



```
@server bin]# ./healthCheck.sh -h
internal config file /com/sterlingcommerce/security/jcae/provider.txt, I
loading it.
mode is being set....true
```

Step 12: Enter **password** for Passphrase.

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Exercise 4.5.2: HealthCheck Command

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Instructions

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Step 13: Click **Enter**, the command displays all the short names to use along with healthCheck.sh, as shown in the screen capture.

```

=====
=====
=====
=====
=====
=====
=====

***** HealthCheck v.1.0.3.1 *****
*****

To select what to run, either go into the properties file or use the command line options to select tests
The available command line options are:
-ALL
    All scanners and all of their tests are turned on
-S abbreviation1 abbreviation2 ...
    The scanners and all of their tests are turned on
Available Scanner Names:
    ConfigScanner: CS
    StaticConfigScanner: SCS
    HostSystemScanner: HSS
    DatabaseScanner: DBS
    DynamicTrackingDataScanner: DTDS
    ScheduleScanner: SS
    PasswordScanner: PS
    SystemProcessScanner: SPS
    BusinessProcessScanner: BPS
    LogScanner: LS
-I testName1 testName2 ...
    The tests are turned on and the scanner(s) they belong to are turned on, if not already
Available Test Names ForCS:
    CS.jettyThreads
    CS.jGroups
    CS.threadStealing
    CS.expiredCertificate
    CS.heapHighWaterMark
    CS.nurseryToHeapRatio
    CS.overrides
    CS.ports
Available Test Names ForSCS:
    SCS.averageCycle
    SCS.cycleTime

```

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Exercise 4.5.2: HealthCheck Command

(Continued)

Instructions

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Angela need to create a report by using business process scanner through command line.

Complete the following steps to run the business process scanner command to generate the report

Step 1: From the Windows image, click PuTTY icon.

Step 2: Select **SI** from Save Session.

Step 3: Click **Open**.

Step 4: Enter **root** for login.

Step 5: Click **Enter**.

Step 6: Enter **root123** for password.

Step 7: Click **Enter**.

Step 8: Type **cd /opt/IBM/SterlingIntegrator/install/bin**

Step 9: Click **Enter**.

Step 10: Type **./healthCheck.sh -S BPS**.

Step 11: Click **Enter**, you see that script loads as shown in the screen capture.

```
bin]# ./healthCheck.sh -S BPS
l config file /com/sterlingcommerce/security/jcae/provider.txt, loading it no
it.
being set....true
ase: -->
```

Step 12: Enter **password** for Passphrase.

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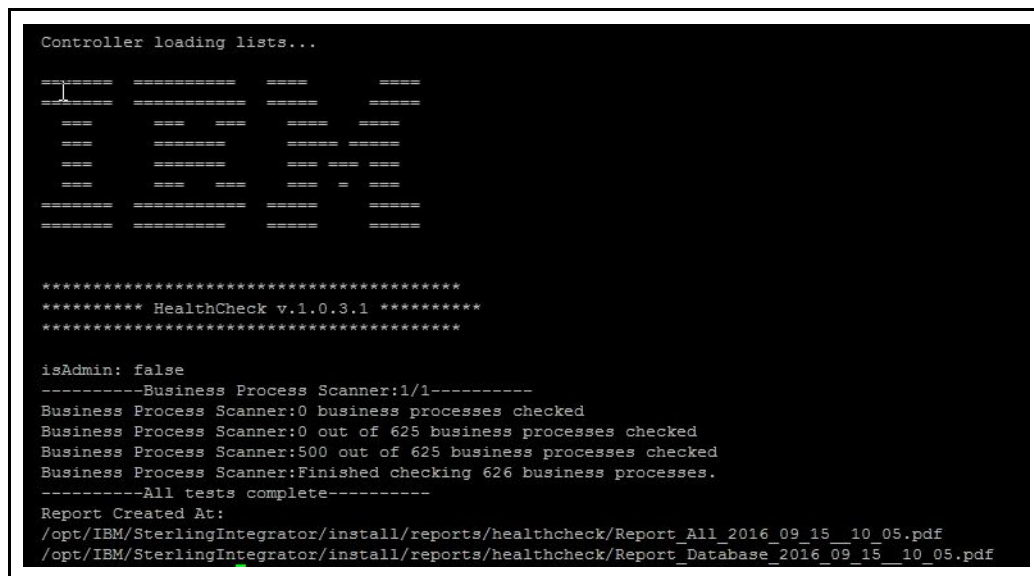
Exercise 4.5.2: HealthCheck Command

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Instructions

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Step 13: Click **Enter**, the command generates the reports, as shown in the screen capture.



```
Controller loading lists...

=====
=====
=====
=====
=====
=====
=====
=====
=====
=====

***** HealthCheck v.1.0.3.1 *****
*****

isAdmin: false
-----Business Process Scanner:1/1-----
Business Process Scanner:0 business processes checked
Business Process Scanner:0 out of 625 business processes checked
Business Process Scanner:500 out of 625 business processes checked
Business Process Scanner:Finished checking 626 business processes.
-----All tests complete-----
Report Created At:
/opt/IBM/SterlingIntegrator/install/reports/healthcheck/Report_All_2016_09_15_10_05.pdf
/opt/IBM/SterlingIntegrator/install/reports/healthcheck/Report_Database_2016_09_15_10_05.pdf
```

Complete the following steps to view the report from the command line:

Step 14: In the PuTTY window, type **cd ../reports/healthcheck**.

Step 15: Click **Enter**

Step 16: Type **ls -ltr**.

Step 17: Click **Enter**, you see the pdf document of business process scanner created.

Complete the following steps to access the report pdf document:

Step 18: From the FileZilla tool, access Sterling B2B Integrator installed directory.

Step 19: The report files are stored in the **/opt/IBM/SterlingIntegrator/install/reports** folder as shown in the screen capture.

Step 20: Select **Report_ALL_XXXX.pdf** file.

Step 21: **Right-click > View/Edit** to view the report in the PDF format.

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HealthCheck Scanner Commands

Introduction

Some of the HealthCheck Scanner commands are depicted in the screen capture:

Table 1. Business Process Scanner - Test Names and Properties. Displays each test in the Business Process Scanner with the command-line short name and property value.

Test Name	Command-Line Short Name	Property
BPMeta	BPS.BPMeta	BusinessProcessScanner.BPMeta
Doc Info	BPS.DocInfo	BusinessProcessScanner.DocInfo
Msg To Child	BPS.msgToChild	BusinessProcessScanner.msgToChild
Sync	BPS.sync	BusinessProcessScanner.sync
This Service	BPS.thisService	BusinessProcessScanner.thisService
X Path	BPS.xPath	BusinessProcessScanner.xPath
Doc To Dom	BPS.docToDom	BusinessProcessScanner.docToDom
Persistence	BPS.Persistence	BusinessProcessScanner.Persistence
Search	BPS.search	BusinessProcessScanner.search

Table 2. Config Scanner - Test Names and Properties. Displays each test in the Config Scanner with the command-line short name and property value.

Test Name	Command-Line Short Name	Property
Active Connections	CS.activeConnections	ConfigScanner.activeConnections
J Groups	CS.jGroups	ConfigScanner.jGroups
Load Balancing	CS.loadBalancing	ConfigScanner.loadBalancing
Ports	CS.ports	ConfigScanner.ports
Thread Stealing	CS.threadStealing	ConfigScanner.threadStealing
Expired Certificate	CS.expiredCertificate	ConfigScanner.expiredCertificate
Heap Water Mark	CS.heapWaterMark	ConfigScanner.heapWaterMark
Jetty Threads	CS.jettyThreads	ConfigScanner.jettyThreads
Nursery To Heap	CS.nurseryToHeap	ConfigScanner.nurseryToHeap
Overrides	CS.overrides	ConfigScanner.overrides

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HealthCheck Scanner Commands

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Introduction

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Table 3. Dynamic Tracking Data Scanner - Test Names and Properties. Displays each test in the Dynamic Tracking Data Scanner with the command-line short name and property value.

Test Name	Command-Line Short Name	Property
Requeue	DTDS.requeue	DynamicTrackingDataScanner.requeue
Size Process Data	DTDS.sizeProcessData	DynamicTrackingDataScanner.sizeProcessData
Halted Process	DTDS.haltedProcess	DynamicTrackingDataScanner.haltedProcess
Interrupted Process	DTDS.interruptedProcess	DynamicTrackingDataScanner.interruptedProcess
Mailbox Old	DTDS.mailboxOld	DynamicTrackingDataScanner.mailboxOld
Mailbox Unextractable	DTDS.mailboxUnextractable	DynamicTrackingDataScanner.mailboxUnextractable
Process Length	DTDS.processLength	DynamicTrackingDataScanner.processLength
Waiting on IO	DTD.waitingOnIO	DynamicTrackingDataScanner.waitingOnIO

Table . Microsoft SQL Server Database Scanner - Test Names and Properties. Displays each test in the Microsoft SQL Server Database Scanner with the command-line short name and property value.

Test Name	Command-Line Short Name	Property
Collation	DBS.mssql.settings.collation	DatabaseScanner.mssql.settings.Collation
Is Auto Create Statistics	DBS.oracle.settings.isAutoCreateStatistics	DatabaseScanner.mssql.settings.isAutoCreateStatistics
Is Auto Update Statistics	DBS.oracle.settings.isAutoUpdateStatistics	DatabaseScanner.mssql.settings.isAutoUpdateStatistics
Page Verify Option	DBS.oracle.settings.page_verify_option	DatabaseScanner.mssql.settings.page_verify_option
Read Committed Snapshot	DBS.oracle.settings.read_committed_snapshot	DatabaseScanner.mssql.settings.read_committed_snapshot

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HealthCheck Scanner Commands

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Introduction

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Table Host System Scanner - Test Names and Properties. Displays each test in the Host System Scanner with the command-line short name and property value.

Test Name	Command-Line Short Name	Property
Dump Files	HSS.dumpFiles	HostSystemScanner.dumpFiles
Cpu Used And IOWait Time	HSS.cpuUsedAndIOWaitTime	HostSystemScanner.cpuUsedAndIOWaitTime
Disk Free	HSS.diskFree	HostSystemScanner.diskFree
Doc Dir Count	HSS.docDirCount	HostSystemScanner.docDirCount
Old Logs	HSS.oldLogs	HostSystemScanner.oldLogs
Thread Count	HSS.threadCount	HostSystemScanner.threadCount
NFS Mount Test	HSS.nfsMount	HostSystemScanner.nfsMount
ULimit	HSS.ulimit	HostSystemScanner.ulimit

Customizing HealthCheck

Introduction

The HealthCheck tool ensures that Sterling B2B Integrator is configured for the production or test system settings to scanner and test property settings.

Extensibility Properties

HealthCheck includes the search and properties tests as extensibility features. The search and properties tests feature can further customized for expanded analysis through extensibility properties.

It is not good practice to changes these settings. If the business requirement does not meet the default settings.

You can customize property settings in two ways:

1. HealthCheck browser user interface
2. HealthCheck property files.

HealthCheck browser user interface

Complete the following steps to edit the values of properties in the user interface:

1. Navigate to the property of the test or scanner.
2. Enter the new values.
3. Run the test or scanner.



Note

The changed values will revert to the original values after a test is run or after you log out of the HealthCheck utility.

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Customizing HealthCheck

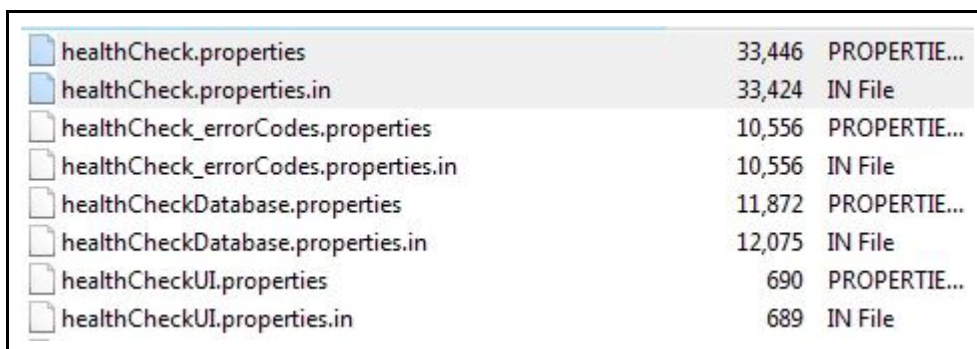
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HealthCheck property files

The HealthCheck utility tool contains all the properties in the healthcheck.properties file. The healthcheck.properties files are stored in <install_dir>/properties folder.

Create or Edit the customer_overrides.properties file to change the properties for customizing the HealthCheck related properties.

The screen capture depicts list of HealthCheck properties file in Sterling B2B Integrator:



healthCheck.properties	33,446	PROPERTIE...
healthCheck.properties.in	33,424	IN File
healthCheck_errorCodes.properties	10,556	PROPERTIE...
healthCheck_errorCodes.properties.in	10,556	IN File
healthCheckDatabase.properties	11,872	PROPERTIE...
healthCheckDatabase.properties.in	12,075	IN File
healthCheckUI.properties	690	PROPERTIE...
healthCheckUI.properties.in	689	IN File

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Exercise 4.5.3: Access HealthCheck property file

Introduction

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Angela accessed all the HealthCheck properties file for better understand of the property for future customization.

Instructions

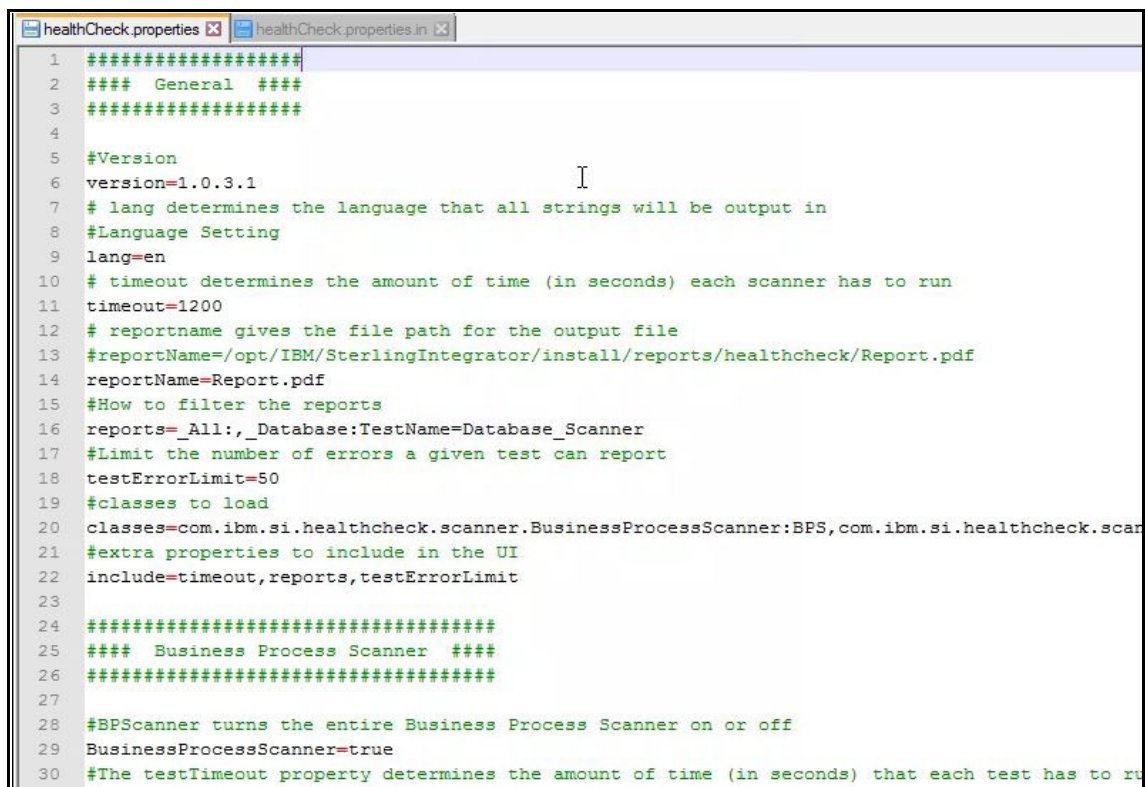
The HealthCheck.properties

The HealthCheck property file is used to enable or disable the scanners to test. By default the scanners are enabled.

Complete the following steps to access healthCheck.properties file:

- Step 1:** From the Windows image, click FileZilla from the desktop to access the Sterling B2B Integrator properties folder.
- Step 2:** From the remote panel, navigate to **/opt/IBM/SterlingIntegrator/install/properties** folder to view the healthcheck related property files.

The screen capture lists few properties of the healthcheck.properties:



```
1 #####
2 ### General ###
3 #####
4
5 #Version
6 version=1.0.3.1
7 # lang determines the language that all strings will be output in
8 #Language Setting
9 lang=en
10 # timeout determines the amount of time (in seconds) each scanner has to run
11 timeout=1200
12 # reportname gives the file path for the output file
13 #reportName=/opt/IBM/SterlingIntegrator/install/reports/healthcheck/Report.pdf
14 reportName=Report.pdf
15 #How to filter the reports
16 reports=_All:,_Database:TestName=Database_Scanner
17 #Limit the number of errors a given test can report
18 testErrorLimit=50
19 #classes to load
20 classes=com.ibm.si.healthcheck.scanner.BusinessProcessScanner:BPS,com.ibm.si.healthcheck.scar
21 #extra properties to include in the UI
22 include=timeout,reports,testErrorLimit
23
24 #####
25 ### Business Process Scanner ###
26 #####
27
28 #BPScanner turns the entire Business Process Scanner on or off
29 BusinessProcessScanner=true
30 #The testTimeout property determines the amount of time (in seconds) that each test has to ru
```

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Exercise 4.5.3: Access HealthCheck property file

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Instructions

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The healthCheck_errorcodes.properties

The healthCheck_errorcodes property file specifies the error codes that are used in different tests such as database, business process, config, system process use. The table lists the code keys for the errors reports and user interface:

Code Key	Description
1 - Warning	is in lowest level, is displayed in green in the reports and the user interface
2 - Error	is in second highest severity level, is displayed in yellow or orange.
3 - Severe	is in highest severity level, is displayed in red.
Other - Unknown	any other error, is displayed in blue

Complete the following steps to access healthCheck_errorcodes.properties file:

Step 1: From the Windows image, click FileZilla from the desktop to access the Sterling B2B Integrator properties folder.

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Exercise 4.5.3: Access HealthCheck property file

(Continued)

Instructions

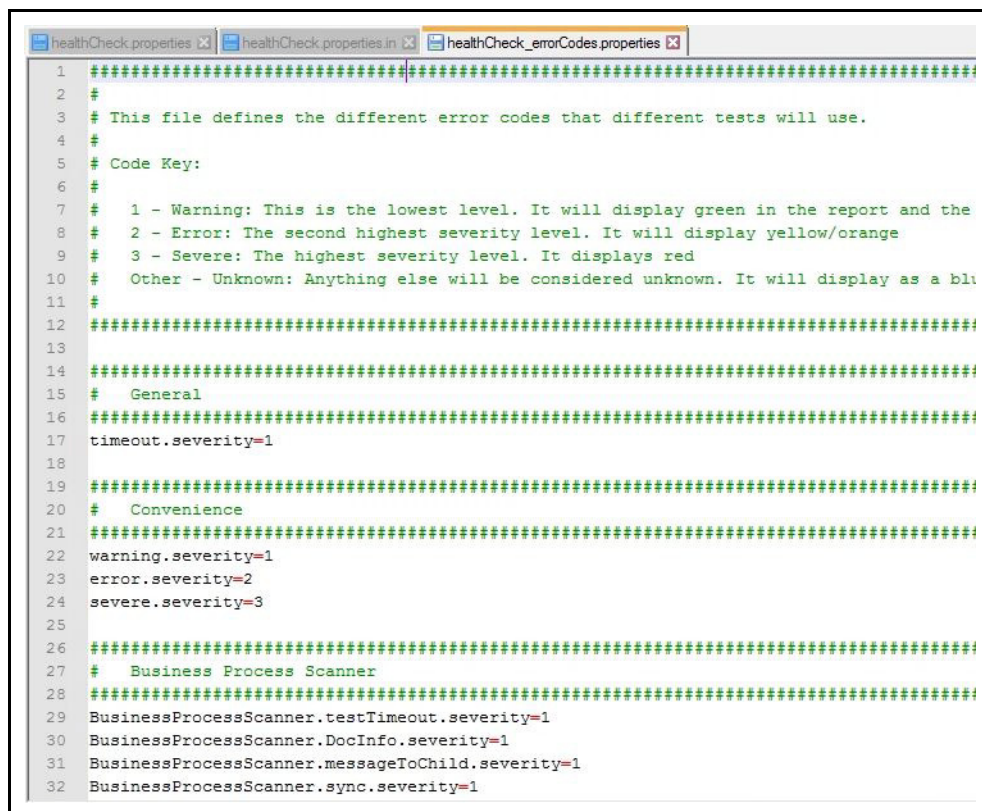
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Step 2: From the remote panel, to `/opt/IBM/SterlingIntegrator/install/properties` folder.

Step 3: Select `healthCheck_errorCodes.properties` file.

Step 4: Right-click > **View/Edit** to view the file.

The screen capture lists few properties of the `healthCheck_errorcodes` file:



```
1 #####
2 #
3 # This file defines the different error codes that different tests will use.
4 #
5 # Code Key:
6 #
7 # 1 - Warning: This is the lowest level. It will display green in the report and the
8 # 2 - Error: The second highest severity level. It will display yellow/orange
9 # 3 - Severe: The highest severity level. It displays red
10 # Other - Unknown: Anything else will be considered unknown. It will display as a bl
11 #
12 #####
13
14 #####
15 # General
16 #####
17 timeout.severity=1
18
19 #####
20 # Convenience
21 #####
22 warning.severity=1
23 error.severity=2
24 severe.severity=3
25
26 #####
27 # Business Process Scanner
28 #####
29 BusinessProcessScanner.testTimeout.severity=1
30 BusinessProcessScanner.DocInfo.severity=1
31 BusinessProcessScanner.messageToChild.severity=1
32 BusinessProcessScanner.sync.severity=1
```

(Continued on next page)

Exercise 4.5.3: Access HealthCheck property file

(Continued)

Instructions

....(Continued)

The healthCheckDatabase.properties

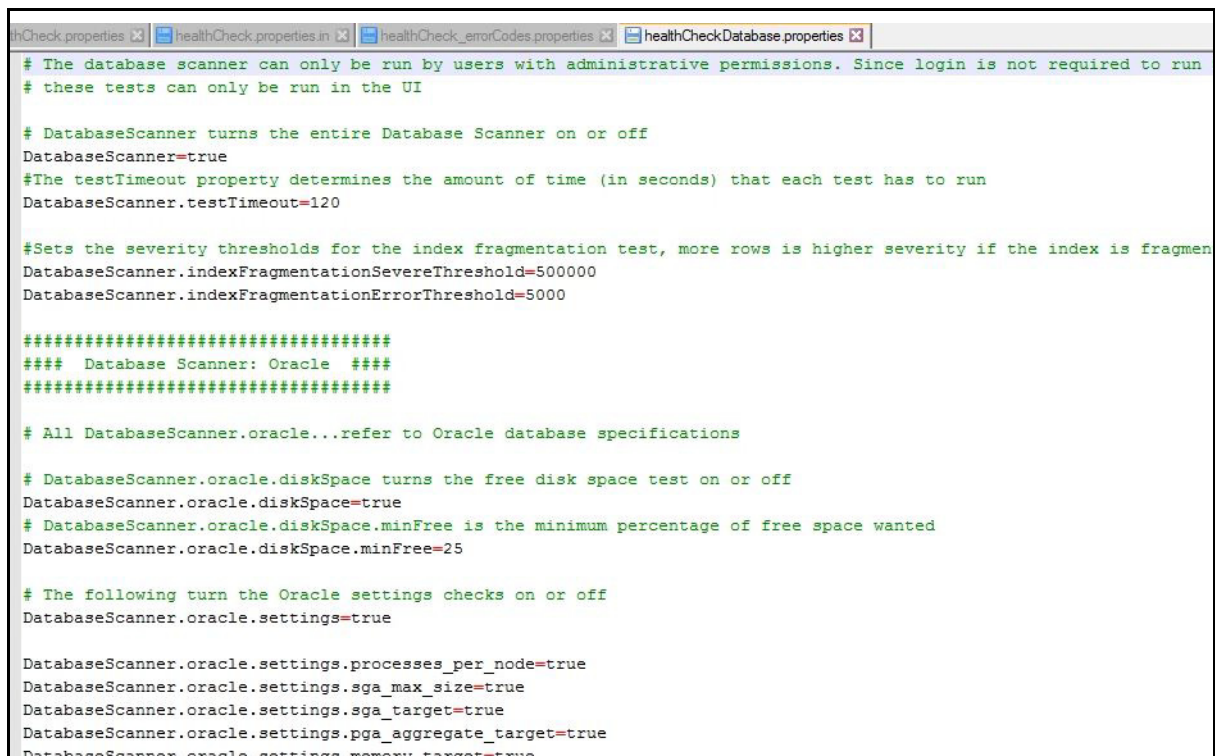
Administrative permission is required to run the healthCheckDatabase property file. The test can be run only through the user interface.

The file contains the customization that is related to database scanner, index compression, index fragmentation, disk space.

Complete the following steps to access healthCheckDatabase.properties file:

- Step 1:** From the Windows image, click FileZilla from the desktop to access the Sterling B2B Integrator properties folder.
- Step 2:** From the remote panel, to **/opt/IBM/SterlingIntegrator/install/properties** folder.
- Step 3:** Select **healthCheckDatabase.properties** file.
- Step 4:** **Right-click > View/Edit** to view the file.

The screen capture lists few properties of the healthCheck database properties file:



```
# The database scanner can only be run by users with administrative permissions. Since login is not required to run
# these tests can only be run in the UI

# DatabaseScanner turns the entire Database Scanner on or off
DatabaseScanner=true
#The testTimeout property determines the amount of time (in seconds) that each test has to run
DatabaseScanner.testTimeout=120

#Sets the severity thresholds for the index fragmentation test, more rows is higher severity if the index is fragmen
DatabaseScanner.indexFragmentationSevereThreshold=500000
DatabaseScanner.indexFragmentationErrorThreshold=5000

#####
### Database Scanner: Oracle ###
#####

# All DatabaseScanner.oracle...refer to Oracle database specifications

# DatabaseScanner.oracle.diskSpace turns the free disk space test on or off
DatabaseScanner.oracle.diskSpace=true
# DatabaseScanner.oracle.diskSpace.minFree is the minimum percentage of free space wanted
DatabaseScanner.oracle.diskSpace.minFree=25

# The following turn the Oracle settings checks on or off
DatabaseScanner.oracle.settings=true

DatabaseScanner.oracle.settings.processes_per_node=true
DatabaseScanner.oracle.settings.sga_max_size=true
DatabaseScanner.oracle.settings.sga_target=true
DatabaseScanner.oracle.settings.pga_aggregate_target=true
DatabaseScanner.oracle.settings.memory_target=true
```

(Continued on next page)

Exercise 4.5.3: Access HealthCheck property file

(Continued)

Instructions

....(Continued)

The healthCheckUI.properties

The healthCheckUI.properties file allows to determine in user interface such as what properties to added, determine to display or hide the high priority items to be displayed at the top or at the bottom in the user interface.

Complete the following steps to access healthCheckUI.properties file:

- Step 1:** From the Windows image, click FileZilla from the desktop to access the Sterling B2B Integrator properties folder.
- Step 2:** From the remote panel, navigate to **/opt/IBM/SterlingIntegrator/install/properties** folder.
- Step 3:** Select **healthcheckUI.properties** file.
- Step 4:** **Right-click > View/Edit** to view the file.

The screen capture lists few properties of the healthCheckUI properties file:

```
#The property name to search for in other property files to determine what properties to add to the UI
ui.includeName=include

#Determine whether or not to include all the properties of known property files
ui.addAll=false

#Determine which additional vendor properties if any should be loaded and displayed in the table for editing
ui.vendorTags=

#Determine which property names will always appear at the top
ui.highPriority=timeout,testTimeout,exclude,reports

#Determine whether high priority items go at the top or at the bottom
# < 0 Goes to the top
# = 0 Does not sort based on priority
# > 0 Goes to the bottom
ui.highPriority.loc=-1

#List of properties that are hidden
ui.hidden=
```



Note

For more details, refer

http://www.ibm.com/support/knowledgecenter/SS3JSW_5.2.0/com.ibm.help.healthcheck.doc/com.ibm.help.healthcheck.reference.doc/healthcheck_properties_overview.html

Recommendation

Introduction

Health Check utility as a diagnostic tool can be used in testing stage, performance tuning stage, and production system, provided the Sterling B2B Integrator administrator is aware of the impact in tuning the properties through health check interface.

To tune the properties of different scanners, refer

http://www.ibm.com/support/knowledgecenter/SS3JSW_5.2.0/com.ibm.help.healthcheck.doc/com.ibm.help.healthcheck.reference.doc/healthcheck_suggested_actions.html

Scripts

Introduction

Scripts are used to run set of instructions to process specific activities in Sterling B2B Integrator system. The scripts are stored in <install_dir>/bin directory. In the training environment, the directory path is /opt/IBM/SterlingIntegrator/install/bin.

Depending on the operating system, you are using for Sterling B2B Integrator, script files have one of the following file extensions:

- .sh for UNIX
- .sh.date for UNIX
- .cmd for Windows
- .cmd.date for Windows

Some scripts can have an .in extension of the entire script name. The .in version of the script is the initial script that is used during the startup process. If you change these scripts, you change the initial scripts permanently. The .in scripts overwrite the changes that are made to the counterpart scripts when you run the setup files script.

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Scripts

(Continued)

Frequently used scripts

The table indicates frequently used scripts by the implementers and administrators are:

File Name	Description
AddLicenseSet	Adds new license information to the database during installation. Deletes the old license file. Saves the old license set in the install_dir/logs directory in a file named similar to oldLicenseSetXMLdate/timeStamp.xml. This file is saved in a subdirectory with a date and time stamp. For example, install_dir/logs/logs_03142016_074334.
deployer	Deploys Sterling B2B Integrator components into the application server environment during installation. Deploys additional components into the application server environment through the InstallService.
deleteVendorJars	Deletes third party jar files.
dump_info	Dumps database information for support purposes after installation.
enccfgs	Encrypts the database and applications server passwords stored in the configuration file. Do not run this script twice; otherwise, error can occur.
hardstop	Stops Sterling B2B Integrator and all processes immediately. The .cmd version stops the Sterling B2B Integrator Windows service and the MySQL Windows service, if MySQL is the database.
InstallService	This script installs the Sterling B2B Integrator packages during installation. This script installs additional packages after installation. Also installs patches downloaded from the Support On Demand website.
install3rdParty	Adds the password library and the jdbc drivers to tmp scripts during installation. Adds third-party components (properties, library and .jar files) to the Sterling B2B Integrator environment after installation.
patchJNLP	Changes the internal IP address used by applications downloaded with Java Web Start.
run.sh	Starts Sterling B2B Integrator. There is no .cmd file equivalent to this script.

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Scripts

(Continued)

Frequently used scripts

....(Continued)

File Name	Description
setupfiles	Applies configuration changes to an installation.
softstop	Shuts down Sterling B2B Integrator, allowing all processes in progress to complete before stopping.
startWindowsService.cmd	Starts the Noapps, Ops, and the Sterling B2B Integrator Windows services. There is no .sh file equivalent to this script.
stopWindowsService.cmd	Stops the Sterling B2B Integrator Windows service. There is no .sh file equivalent to this script.
tee.exe	Used for logging. There is no .sh file equivalent to this file. Do not use or edit this file.
tmp.cmd	Used by many scripts to obtain Sterling B2B Integrator environment information. Do not use or edit this script. There is no .sh file equivalent to this script.
workflowLauncher	Starts business processes from the command line.

For more scripts, view **/opt/IBM/SterlingIntegrator/install/bin** directory through FileZilla tool in the lab environment.

For more details about the scripts refer:

http://www.ibm.com/support/knowledgecenter/SS3JSW_5.2.0/com.ibm.help.bin_directory.doc/SI_binDir_files.html

Exercise 4.5.4: Run Business Process

Introduction

Alice starts a business process execution from the command line, which in turn will help in real time tuning in future.

Instructions

Complete the following steps to run the business process through command line:

Step 1: From the windows image, click **FileZilla** tool.

Step 2: Click drop down icon next to **QuickConnect**.

Step 3: Select **sftp://root@192.168.40.100** to connect to the server system.

Step 4: From the Local site panel, copy InventoryOf3.xml file from C:\6F89G\Labfiles to Remote site panel, navigate to **/home** directory.

Step 5: From the windows image, click PuTTY icon to access the server system.

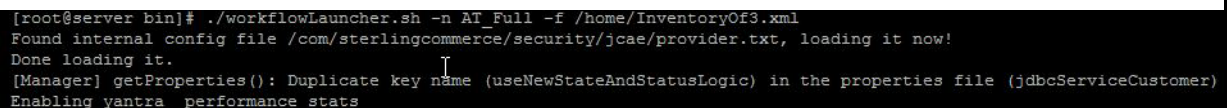
Step 6: Enter **root** as Login as and **root123** as Password.

Step 7: Change the directory to bin directory as:

```
cd /opt/IBM/SterlingIntegrator/install/bin
```

Step 8: Enter **./workflowLauncher.sh -n AT_Full -f /home/InventoryOf3.xml**.

Step 9: Click **Enter**, to view result, as shown in the screen capture.



```
[root@server bin]# ./workflowLauncher.sh -n AT_Full -f /home/InventoryOf3.xml
Found internal config file /com/sterlingcommerce/security/jcae/provider.txt, loading it now!
Done loading it.
[Manager] getProperties(): Duplicate key name (useNewStateAndStatusLogic) in the properties file (jdbcServiceCustomer)
Enabling yantra performance stats
```

Step 10: Alternatively, you can view the process in **Business Process > Monitor > Current Process** of dashboard.

Exercise 4.5.5: Create Database Details

Introduction

Alice need to extract the database information from the Sterling B2B Integrator system for the support team to analyze the database details of the system.

Instructions

Complete the following steps to run dumpinfo script:

Step 1: From the windows image, click PuTTY icon to access the server system.

Step 2: Enter **root** as Login as and **root123** as Password.

Step 3: Change the directory to bin directory as:

```
cd /opt/IBM/SterlingIntegrator/install/bin
```

Step 4: Enter **./dump_info.sh > dump_info.log** .

Step 5: Click **Enter**. A file with name “dump_info.log will be created in the /opt/IBM/SterlingIntegrator/install/bin directory.

Step 6: Type **ls dump_info***.

Step 7: Click **Enter** to view the dump_info.log file.

Step 8: Type **vi dump_info.log**.

Step 9: Click **Enter** to view the content of the file.

Step 10: Enter Esc **:q!** to exit from vi tool.

Lesson review

What you have been able to do

After completing this lesson, you should have been able to:

- Explain about HealthCheck tool.
 - Discuss the importance of HealthCheck tool.
 - Explain the procedure to install HealthCheck tool.
 - Describe the steps to customize HealthCheck tool.
 - Explain the importance of the scripts in Sterling B2B Integrator.
-